RideShield Prototype Parts List

Oct 11, 2026

Part	Qty	Purpose
ESP32-S3 N16R8	1	Main controller
ESP32-S3-CAM	2	Camera boards for blind spot and road view; skip separate cameras if included
MPU6050	1	Head motion and crash detection
ADXL375	1	Hard impacts and crash severity
VL53L0X	1	Helmet-worn check
Reed switch + magnet	1	Chin strap check (a microswitch also works)
OP0062	1	Sleep detection
MQ-3B	1	Alcohol check
GNSS module + antenna	1	Location and speed
4G LTE module + antenna + SIM	1	Emergency alerts
microSD module + 16–32 GB card	1	Black box log
SOS button	1	Manual emergency alert
Cancel button	1	Cancel a false alarm
Active buzzer	1	Sound warning
Vibration motor	1	Haptic warning
LED	3	Status and warning lights
LiPo battery	1	Powers the helmet
LiPo charger	1	Charges the battery